Credit Card Fraud Detection Using Machine Learning: A Comparative Study of Classification Models

SECTION I.

Abstract

Financial institutions play a central role in economic growth by facilitating the flow of funds between depositors, investors, and borrowers. As digital transactions increase, these institutions face growing exposure to financial fraud, making effective risk mitigation strategies essential. Timely and accurate fraud prediction is therefore critical for maintaining operational efficiency and trust.

This study evaluates multiple machine learning algorithms implemented in Weka, including J48, IB1, OneR, Naive Bayes, and AdaBoostM1, within a FinTech case scenario. Results show that, on imbalanced data, most models achieve over 99% accuracy while failing to detect fraudulent transactions. In contrast, Naive Bayes detects all fraud but generates excessive false positives. A resampling experiment demonstrates that balancing the dataset reduces accuracy but significantly improves model validity, as reflected by higher Kappa values.

The findings highlight that accuracy is not a reliable metric in imbalanced datasets and emphasize the importance of preprocessing techniques. Overall, the study concludes that data distribution has a greater impact on model performance than algorithm selection, reinforcing the need for imbalance-aware approaches in real-world fraud detection systems.

SECTION II.

Introduction

Fraudulent transactions and the systems designed to detect them have evolved in parallel over time. In the current Internet-driven economy, digital financial systems have significantly increased exposure to fraudulent activity. Reports indicate that transaction fraud resulted in losses of approximately $28 billion in 2019, rising to $30 billion in 2020 and exceeding $32 billion in 2021, with projections reaching around $34 billion in 2022. This upward trend highlights the growing scale and sophistication of fraudulent activities within digital financial ecosystems.

This study considers a FinTech company, MGH *Ltd*., which provides digital financial services including online payment processing, peer-to-peer transactions, and small-scale investment management. The platform operates entirely online, serving approximately 50,000 active users and processing between 20,000 and 30,000 transactions daily. The company generates revenue primarily through transaction fees and investment-related services, making transaction integrity essential to its business model.

Due to the high transaction volume and real-time nature of operations, manual fraud detection is not feasible. Instead, the company relies on automated systems to identify suspicious activities and prevent fraudulent transactions. A key challenge faced by MGH *Ltd*.. is the highly imbalanced

nature of transaction data, where fraudulent cases represent only a small fraction of total activity. This imbalance makes it difficult to detect fraud accurately without incorrectly flagging legitimate transactions, which can negatively impact customer experience and trust.

In this context, machine learning models play a crucial role in enabling efficient and scalable fraud detection. This study aims to evaluate the performance of multiple classification algorithms and analyse how class imbalance affects their effectiveness. Additionally, the impact of resampling techniques is examined to determine whether balancing the dataset improves model reliability and practical applicability.

SECTION III.
Review Methodology

Two datasets were used in this study to evaluate model performance under different data conditions. The primary dataset, sourced from Kaggle, consists of real-world financial transactions and is characterized by a high class imbalance, where fraudulent cases represent only a small proportion of the data. This dataset was selected because it reflects realistic fraud detection scenarios, allowing the evaluation of how machine learning models behave in practical environments.

A second dataset was used in a resampled (balanced) form to address the limitations of the original dataset. This allowed for controlled comparison and enabled the assessment of how class distribution impacts model performance.

The use of both imbalanced and balanced datasets provides a more comprehensive evaluation of the algorithms, supporting a deeper understanding of their strengths and limitations in fraud detection tasks.

3.1. Materials and Methods

The development of the prediction model followed a structured workflow consisting of data preprocessing, model training, and evaluation. This approach was selected to ensure that the models could be applied effectively within the constraints of the dataset and the experimental environment.

Data preprocessing was necessary to address formatting inconsistencies, convert attribute types, and reduce dataset size to ensure compatibility with Weka. These steps are critical in fraud detection, where data quality directly affects model performance (Kulatilleke, 2022).

A key aspect of the approach was the use of 10-fold cross-validation, which was chosen to provide a reliable and unbiased estimate of model performance by reducing variance associated with a single train-test split. This is particularly important in fraud detection tasks characterized by imbalanced datasets (Thennakoon et al., 2019).

Additionally, a resampling technique was applied to create a balanced version of the dataset. This allows for a comparative analysis between imbalanced and balanced data conditions, as recommended in prior research addressing class imbalance challenges (Carcillo et al., 2021).

Overall, this approach ensures that the evaluation is both methodologically sound and aligned with best practices in fraud detection modelling.

3.2 .Machine Learning Classifiers

Using the dataset prepared in the previous stages, a set of machine learning classifiers was implemented in Weka to evaluate their performance in fraud detection. The selected models included J48 (Decision Tree), IB1 (k-Nearest Neighbour), OneR, Naive Bayes, and AdaBoostM1. These algorithms were chosen to represent different classification paradigms, including rule-based, probabilistic, instance-based, and ensemble learning approaches.

The experimental analysis was conducted under two different data conditions. First, the original imbalanced dataset (approximately 20,000 instances) was used to train and evaluate the classifiers. All models were implemented using their default parameters to provide a baseline comparison and to assess their behaviour under realistic fraud detection conditions.

In the second experiment, a resampling technique was applied to generate a more balanced dataset. The same classifiers were then trained and evaluated again using identical settings. The purpose of this comparison was to analyse the impact of class distribution on model performance and to determine whether balancing the dataset improves classification effectiveness.

For both experiments, model performance was evaluated using 10-fold cross-validation, ensuring a robust and unbiased assessment by reducing the variability associated with a single train-test split. This approach allowed for consistent comparison across models and datasets, providing insight into the strengths and limitations of each algorithm in fraud detection tasks.

3.3. Performance Evaluation

The performance of the machine learning models was evaluated using multiple classification metrics to provide a comprehensive assessment beyond overall accuracy. This is particularly important in fraud detection, where class imbalance can lead to misleading results if only accuracy is considered (Thennakoon et al., 2019).

The primary evaluation method used was **10-fold cross-validation**, implemented in Weka. This approach involves dividing the dataset into ten equal subsets, where each subset is used once as a test set while the remaining subsets are used for training. This process reduces variance and provides a more reliable estimate of model performance.

The following performance metrics were analysed:

Accuracy: Measures the proportion of correctly classified instances. However, in imbalanced datasets, high accuracy may not reflect true model effectiveness.
Precision: Indicates the proportion of correctly predicted fraud cases among all predicted fraud instances, reflecting the model's ability to minimize false positives.
Recall (Sensitivity): Measures the ability of the model to correctly identify fraudulent transactions, which is critical in fraud detection tasks.
F-Measure: Provides a balance between precision and recall, offering a more comprehensive evaluation of classification performance.
Kappa Statistic: Measures agreement between predicted and actual classifications beyond chance, making it particularly useful in imbalanced scenarios.

In addition to these metrics, **confusion matrices** were used to analyse classification outcomes in detail, allowing for identification of true positives, false positives, true negatives, and false negatives. This provides deeper insight into model behaviour, particularly in distinguishing between legitimate and fraudulent transactions.

Overall, this evaluation framework ensures a robust and realistic assessment of model performance in the context of fraud detection.

SECTION IV.

Results Analysis

The performance of the machine learning models was evaluated using 10-fold cross-validation in Weka. The results reveal significant differences in model behaviour, particularly due to the impact of class imbalance.

For the original imbalanced dataset, most classifiers, including J48, IB1, OneR, and AdaBoostM1, achieved very high accuracy values exceeding 99%. However, confusion matrix analysis shows that these models classified all instances as non-fraud (class 0), resulting in zero detection of fraudulent transactions (recall = 0). This indicates a strong bias toward the majority class, a common limitation of standard classifiers in imbalanced datasets (Kulatilleke, 2022).

In contrast, the Naive Bayes classifier exhibited the opposite behaviour. It achieved perfect recall (1.000), successfully identifying all fraudulent transactions. However, it also produced a very high number of false positives, incorrectly classifying a large proportion of legitimate transactions as fraud. This resulted in extremely low precision, highlighting the trade-off between detecting minority classes and maintaining overall prediction accuracy (Sailusha et al., 2020).

To further investigate the effect of class imbalance, a second experiment was conducted using a resampled dataset with a more balanced class distribution. In this case, overall accuracy decreased to approximately 59%, but the Kappa statistic increased to around 0.59, indicating a significant improvement in model reliability. This aligns with prior research showing that balancing techniques can improve classification performance in fraud detection tasks (Carcillo et al., 2021).

These findings confirm that class imbalance is the primary factor influencing model performance. While the imbalanced dataset produced high accuracy, it failed to support effective fraud detection. The resampled dataset, although yielding lower accuracy, provided a more realistic evaluation of model capability.

Overall, the testing results demonstrate that accuracy alone is not a sufficient metric for evaluating fraud detection models. Instead, metrics such as recall, precision, and Kappa provide a more accurate representation of performance, particularly in imbalanced scenarios (Thennakoon et al., 2019). This highlights the importance of appropriate evaluation strategies in the development of reliable fraud detection systems.

4.1 Comparison Tables

Table 1. Model Performance on Imbalanced Dataset

Model	Accuracy (%)	Precision	Recall (Fraud)	F1-Score	Kappa
J48	99.57	-	0,000	-	0,000
IB1 (k-NN)	99.53	0,992	0,000	0,993	-0,0007
OneR	99.57	-	0,000	-	0,000
Naive Bayes	4.46	0,004	1,000	0,009	0,0004
AdaBoostM1	99.57	-	0,000	-	0,000

Table 1. Performance comparison of machine learning models on the imbalanced dataset. Despite high accuracy values, most models fail to detect fraudulent transactions, as indicated by zero recall.

Table 2. Model Performance on Resampled (Balanced) Dataset

Model	Accuracy	Kappa
J48	59.63	0.5929
IB1 (k-NN)	59.39	0.5915
OneR	59.24	0.5909
AdaBoostM1	1.11	0.0018

Table 2. Performance of machine learning models on the resampled dataset. Lower accuracy is observed; however, higher Kappa values indicate improved model reliability and balanced classification.

SECTION V.

Discussion

The approach developed in this study can be applied directly to real-world fraud detection systems in FinTech environments. The results show that standard classifiers achieve high accuracy on imbalanced datasets while failing to detect fraudulent transactions, a limitation widely reported in the literature (Mittal and Tyagi, 2019).

This highlights the importance of incorporating preprocessing techniques such as resampling, as relying on raw data alone can lead to ineffective models. Previous studies confirm that handling class imbalance is essential for improving fraud detection performance (Carcillo et al., 2021).

The results also demonstrate that no single algorithm is sufficient. While some models detect fraud effectively, they may produce excessive false positives, indicating the need for hybrid or ensemble approaches, as supported by prior research (Karthik et al., 2022; Taha and Malebary, 2020).

Overall, the findings emphasize that effective fraud detection requires a combination of appropriate algorithms, data preprocessing, and robust evaluation methods to ensure reliable performance in real-world applications.

SECTION VI.

Conclusion

Machine learning offers significant potential for improving fraud detection in financial systems. However, this study demonstrates that model performance is heavily influenced by data characteristics, particularly class imbalance.

The results show that high accuracy does not necessarily indicate effective fraud detection, as models may fail to identify fraudulent transactions. The use of resampling techniques improves model reliability by enabling more balanced classification.

Overall, effective fraud detection requires a combination of appropriate algorithms, data preprocessing, and robust evaluation metrics. Future work may explore more advanced and hybrid approaches to further enhance detection performance.

SECTION VII.

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Datasets and Online Sources

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SECTION VIII.

Appendix A: WEKA Classification Outputs

This appendix provides supporting WEKA outputs corresponding to the results discussed in Section 3 and 4.

Figure A1. J48 – Confusion Matrix (Imbalanced Dataset)

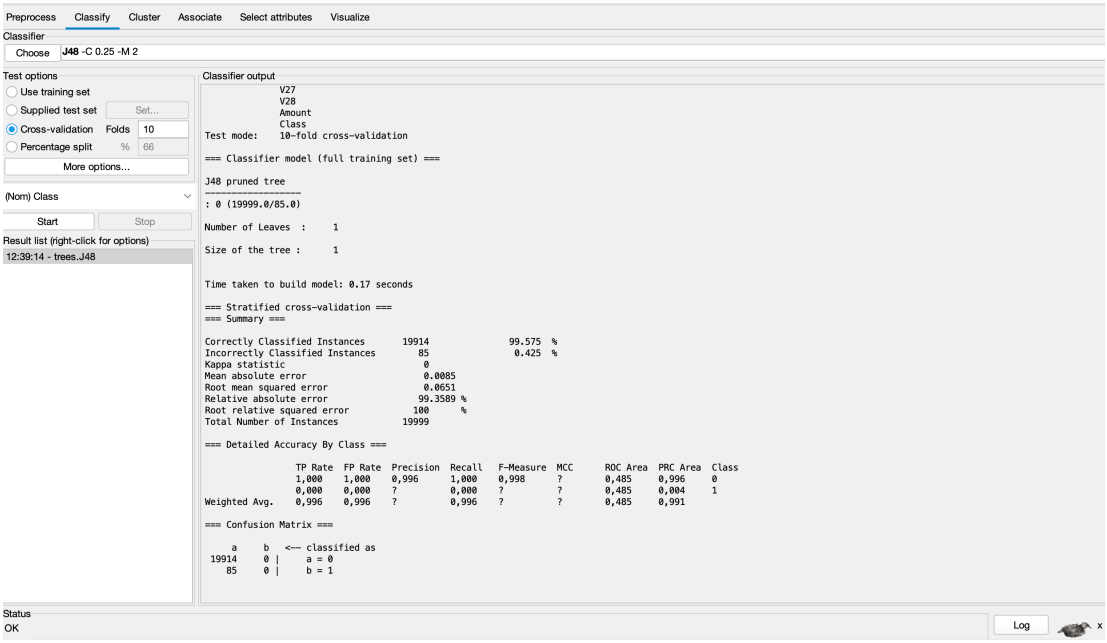


Figure A2. IB1 – Confusion Matrix (Imbalanced Dataset)

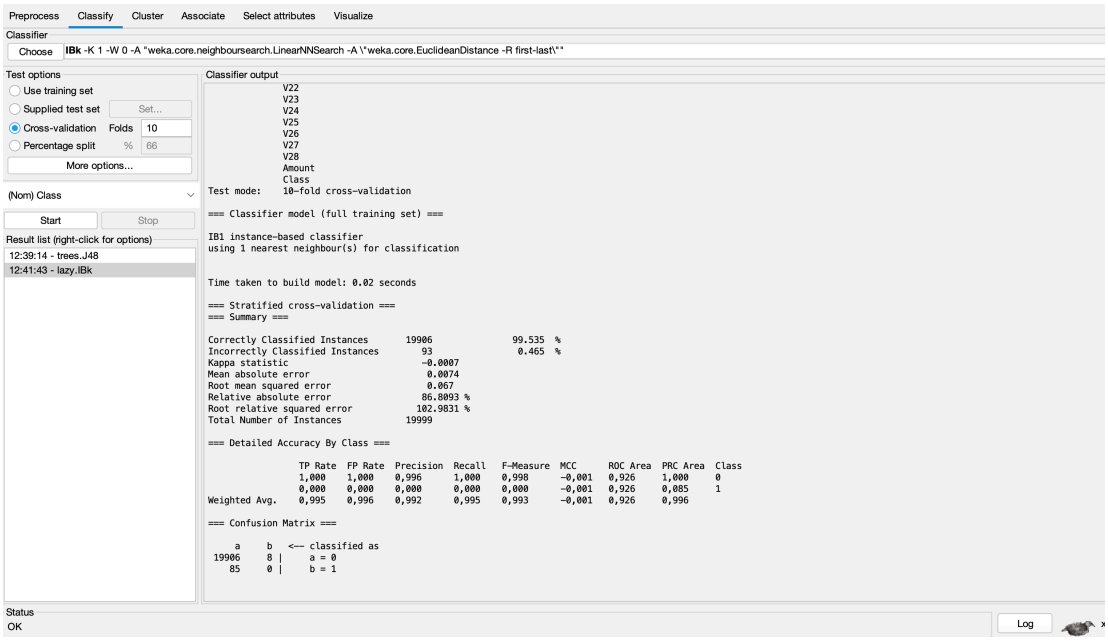


Figure A3. Naive Bayes – Confusion Matrix (Imbalanced Dataset)

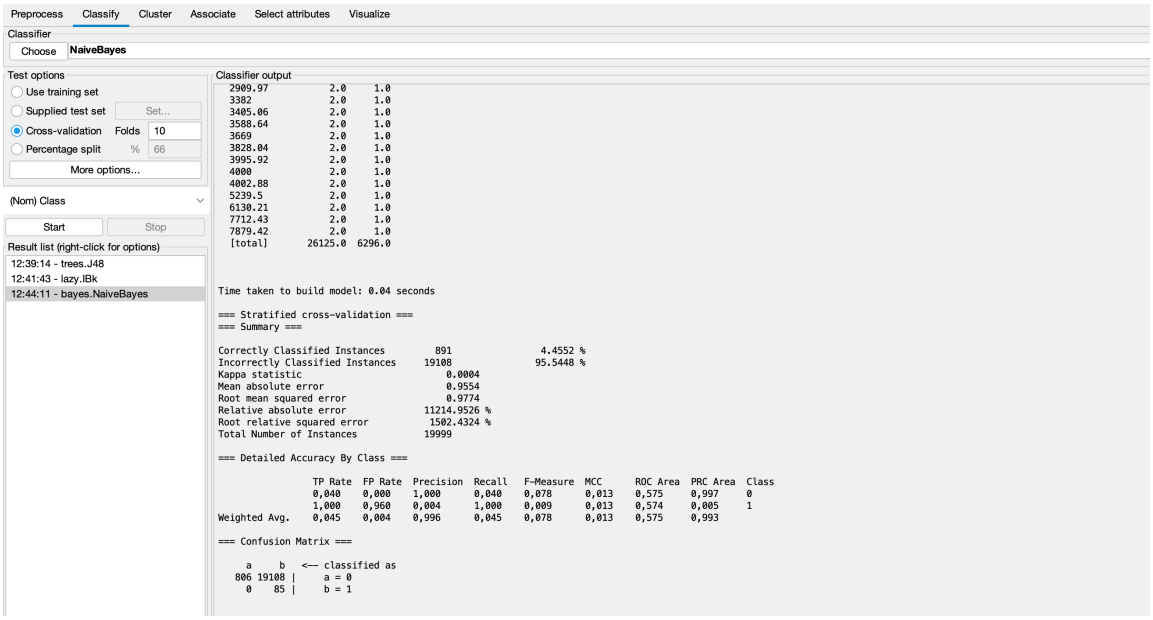


Figure A4. AdaBoostM1 – Confusion Matrix (Imbalanced Dataset)

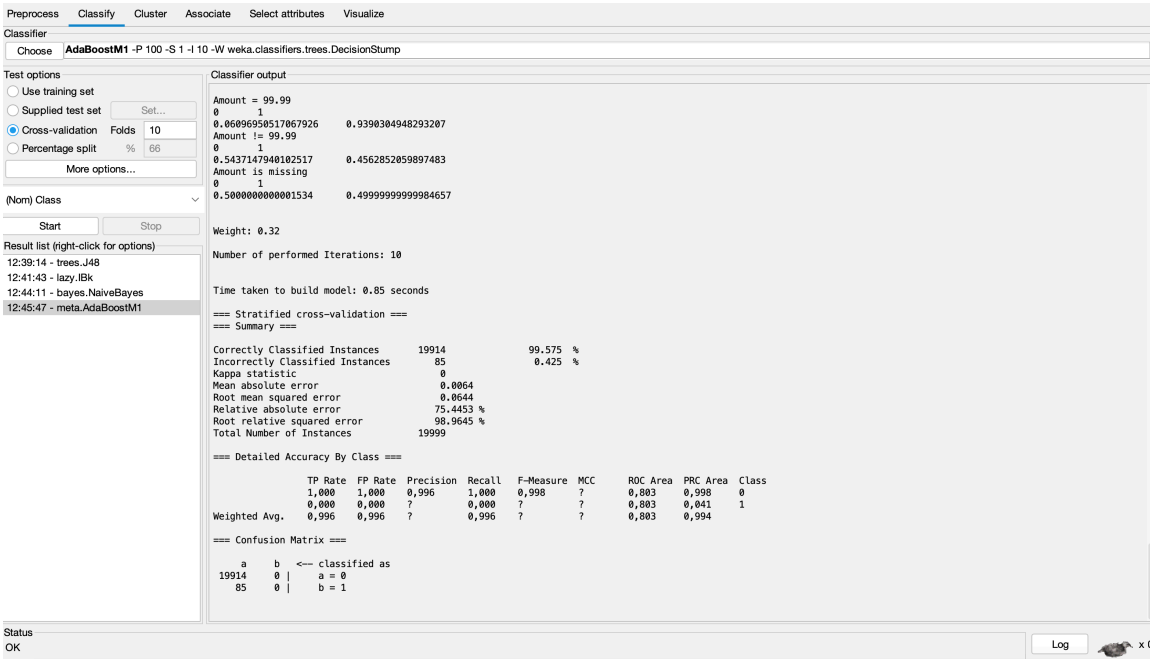


Figure A5. J48 – (Resampled Dataset)

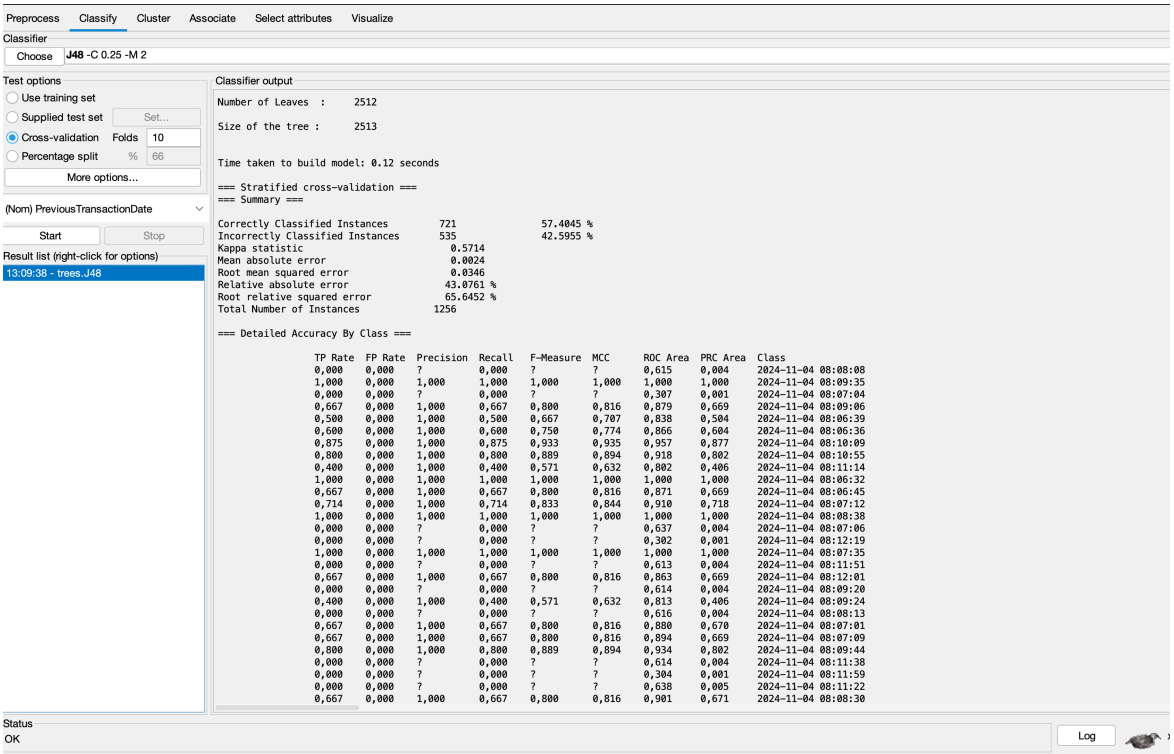


Figure A6. IB1 – (Resampled Dataset)

